

Lead Scoring Model

To build a lead scoring model, we need to observe the final outcome of each deal (Closed Won or Closed Lost). In the dataset, there are 9,908 deals with a closed status, and all of them reach at least the Opportunity stage. This means that no deals are marked as lost before this phase.

As a result, there is no visibility on leads that drop off earlier in the funnel (e.g., at MQL or SQL stages). This introduces a survivorship bias, as only more advanced deals are observed in the final outcome, and limits the ability to build accurate models for early-stage conversion.

From a modelling perspective, the most natural point to apply lead scoring would be the Opportunity stage. However, the funnel analysis revealed a strong data leakage issue: the variable demo score almost perfectly determines the final outcome at this stage. As a result, a model built at this point would not provide real predictive value.

To address this, I designed the model to rely only on features that are available early in the funnel, such as deal characteristics (e.g., industry) and acquisition channel (e.g., deal source). These variables are already known from the MQL/SQL stages and remain informative later in the process.

This approach ensures that the model can be applied consistently from the MQL stage onward, where it can effectively support lead prioritisation. **However, it is important to note that the model is trained on deals that have already progressed to later stages of the funnel. As a result, predictions for early-stage leads may be less accurate, since the training data does not fully represent all initial leads.**

Before the MQL stage, the progression appears to be driven mostly by deterministic rules, making a predictive model less useful.

Model Performance

Before training the model, I performed the following data cleaning steps:

- Removed records with inconsistent future dates
- Excluded deals with lost reason "disqualified"

- Removed null values while tracking the number of dropped observations

To build the lead scoring model, I focused on features that are available early in the funnel. The most relevant features identified are:

- Deal source
- Deal industry
- Contact role
- Company state

Model Comparison: Logistic Regression vs XGBoost

To evaluate whether introducing non-linearities improves performance, I compared a logistic regression baseline with a gradient boosting model (XGBoost).

Evaluation Metrics

- **ROC-AUC:** measures how well the model distinguishes between won and lost deals across all thresholds. Higher values indicate better discrimination.
- **Top-20% conversion rate:** measures how many of the highest-scored leads actually convert. This reflects how useful the model is for prioritisation.
- **Lift@20%:** compares the conversion rate of the top 20% of leads to the overall average. It quantifies how much better the model performs compared to random selection.
- **Missed won rate:** measures the proportion of won deals that are not identified as high-quality leads. This captures the risk of missing valuable opportunities.

Performance Comparison

Logistic regression consistently outperforms XGBoost across all key metrics:

Metric	Logistic Regression	XGBoost
ROC-AUC	● 0.77	● 0.73
Top-20% Conversion Rate	● 80.0%	● 75.1%
Lift@20%	● 1.91x	● 1.79x

Metric	Logistic Regression	XGBoost
Missed Won Rate	 39.6%	 40.9%

Both models show strong ranking behaviour, with conversion rates decreasing consistently across score buckets. However, logistic regression achieves a clearer separation between high- and low-quality leads.

Implications

Model Usage: Ranking vs Filtering

The models show strong ranking performance, with high conversion rates among top-scored leads. However, it also presents a relatively high missed won rate, meaning that a significant portion of good leads receive lower scores. Thus:

- The model is well suited for prioritisation, as it effectively ranks leads by quality
- The model is **not suitable for strict filtering**, as discarding low-scored leads would result in losing valuable opportunities

The model can support the team by:

- Prioritising top-scored leads for immediate follow-up to reduce early-stage drop-off
- increasing persistence (e.g., number of calls) on high-scored leads, which has been shown to improve conversion rates

Model Monitoring

Apart from the metrics detailed above, the impact of the model can be monitored through business KPIs such as conversion rate over time and the duration of the sales process, to assess whether better prioritisation improves efficiency and accelerates deal closure.

Risk and Limitations

- Changes in acquisition channels, industries, or customer profiles may reduce model performance over time
- New or unseen categories (e.g., new deal sources or industries) may not be properly handled by the model

- The model is trained only on deals that reached later stages of the funnel, which may limit its accuracy for early-stage leads